

1. Consider wine dataset with Target Variable: Wine class (0, 1, 2) and perform the following:
 - (a) Use PCA for dimensionality reduction.
 - (b) Split the dataset into training and testing sets in 80:20 ratio.
 - (c) Train and compare Logistic Regression, Naive Bayes, and Decision Tree classifiers.
 - (d) Evaluate and compare their performance using various metrics (For each classifier, report accuracy, precision, recall, F1 score, R^2 , and ROC-AUC score also visualize the confusion matrix for each model.).
 - (e) Conduct hypothesis testing to determine if there are significant differences between classifier performances (f-test, and t-test)(Inbuilt functions can be used).

The wine dataset is readily available in the scikit-learn package and can be loaded using the code below

```
from sklearn import datasets
iris = datasets.load_wine()
```